

Comparative Study of Classification Algorithms: Poker Hand Classification

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1 Introduction and Motivation

Supervised learning constitutes the backbone of many Machine Learning applications. The choice of algorithm is not universal, but depends on the nature of the problem, the characteristics of the dataset, and the intrinsic assumptions of the model. The primary objective of this work is twofold:

- Implement and tune models ranging from probabilistic approaches (Naive Bayes) to linear (Softmax Regression) and tree-based methods (Decision Tree, Random Forest).
- Conduct a rigorous comparative analysis to understand how different algorithmic hypotheses influence the generalization capacity and performance on the selected dataset.

The problem of **Poker Hand Classification** was chosen, a complex and representative task due to its multi-class nature (10 classes) and the notable imbalance between classes.

2 Dataset Description and Preprocessing

2.1 Dataset

The dataset used is the **Poker Hand Classification** from the UCI Machine Learning Repository, containing a total of 1,025,010 samples (well over the 1,000 required). Each sample describes a five-card poker hand through 10 attributes, grouped into pairs (S1, C1), (S2, C2), ..., (S5, C5), where 'S' indicates the suit (1-4) and 'C' the card rank (1-13).

The classification target consists of 10 unique classes, representing the type of poker hand, from *Nothing in hand* (Class 0) to *Royal Flush* (Class 9). A **strong class imbalance** was observed: Class 0 constitutes over 50% of the dataset, while the rarest hands like Class 9 represent less than 0.001%.

2.2 Preprocessing and Methodology

The original dataset did not present missing or noisy data. The implemented preprocessing steps were:

- **Stratified Splitting:** The data was divided into Training (70%), Validation (15%), and Test (15%) sets. Stratification is crucial in the presence of imbalance to ensure that each set maintains the same class distribution.
- **Normalization:** The *StandardScaler* was applied to standardize the features, a fundamental step for distance- or gradient-based models, such as Softmax Regression.

3 Models

3.1 Model Implementation and Hyperparameter Tuning

For all models, **Cross-Validation** (Grid Search with Stratified K-Fold) was performed on the training set for hyperparameter tuning, using Accuracy as the selection metric. The Weighted F1 Score, more robust than simple Accuracy in the presence of imbalance, was then used as the main metric to compare the models.

- **Naive Bayes (Gaussian):** A probabilistic model that assumes the conditional independence of features.
Best Params: $\{\text{'var_smoothing'} : 1e - 09\}$
- **Softmax Regression:** An extension of Logistic Regression for multi-class problems. Used with the option *'class_weight': 'balanced'* to mitigate the effect of imbalance.
Best Params: $\{\text{'C'} : 0.01, \text{'class_weight'} : \text{'balanced'}\}$
- **Decision Tree:** A non-linear model that partitions the feature space into rectangular regions.
Best Params: $\{\text{'class_weight'} : \text{'balanced'}, \text{'max_depth'} : \text{None (unconstrained)}\}$
- **Random Forest:** An Ensemble of Decision Trees trained on random subsets of data and features.
Best Params: $\{\text{'class_weight'} : \text{'balanced'}, \text{'max_depth'} : 20, \text{'n_estimators'} : 200\}$

4 Results and Analysis

4.1 Model Performance Summary

Model	CV Score (Train)	Validation Accuracy	Test Accuracy	Test Precision	Test Recall	Test F1	Training Time (s)
Naive Bayes	0.106653	0.013134	0.013021	0.446656	0.013021	0.021937	1.594389
Softmax Regression	0.144214	0.093106	0.094548	0.461648	0.094548	0.141462	1860.530950
Decision Tree	0.651245	0.664280	0.664772	0.665115	0.664772	0.664917	233.678683
Random Forest	0.743605	0.754901	0.757050	0.745226	0.757050	0.732397	12085.768614

Table 1: Model Comparison Summary (Test Set)

4.2 Comparative Analysis

4.2.1 Which Models Performed Best and Why

The **Random Forest** is the model with the superior performance across all key metrics (Accuracy, Precision, Recall, F1 Score). This is due to two main factors:

1. **Non-linearity and Feature Interaction:** The poker hand classification problem is highly non-linear (e.g., the interaction between suits and ranks defines a hierarchical scale of hands). Tree-based models, and particularly the Random Forest ensemble, are intrinsically capable of capturing complex interactions and non-linear decision boundaries, unlike Softmax Regression.
2. **Variance Reduction (Ensemble):** As an ensemble model, Random Forest aggregates the predictions of individual trees, which tend to have high variance (overfitting). The process of voting (or averaging) significantly reduces variance, improving generalization capacity.

The **Decision Tree** is ranked second but shows a performance gap compared to the Random Forest, highlighting the value of the ensemble approach.

4.2.2 Influence of Model Assumptions

The assumptions of each model had a direct impact on performance:

- **Naive Bayes:** Its extremely low performance (Accuracy = 1.30%) is a direct consequence of its assumption of **conditional independence of features**. In poker, the values and suits of the five cards are highly interdependent in defining a hand (e.g., 5 cards of the same suit = flush). This assumption was spectacularly violated. In addition, training with balanced sample weights pushed the model to over-predict the rare classes, assigning most hands of Class 0 and 1 to combinations such as straight, flush or full house. Accuracy therefore falls far below the share of the majority class (*Nothing in hand*).
- **Softmax Regression:** By assuming **linear separability** between classes (after the logit transformation), the model failed to capture the complex decision boundaries. Although the use of *class_weight='balanced'* improved its F1 Score compared to Naive Bayes, it could not overcome its fundamental limitation on non-linear data.

4.2.3 Observations on the Overfitting Trade-off

Overfitting is the gap between performance on the training/validation set and the test set.

- **Random Forest:** Shows the best generalization. Its CV Score was 0.7436, and the Test Accuracy is 0.7571. This $\Delta\text{Accuracy} \approx 0.01$ shows that hyperparameter optimization (specifically the constraint on maximum depth `max_depth = 20` and sampling) effectively mitigated the overfitting tendency of individual trees.
- **Decision Tree:** Has a CV Score of 0.6512 and a Test Accuracy of 0.6648. Although the difference is small, the unconstrained tree has a greater theoretical potential for having memorized the training data compared to the Random Forest.

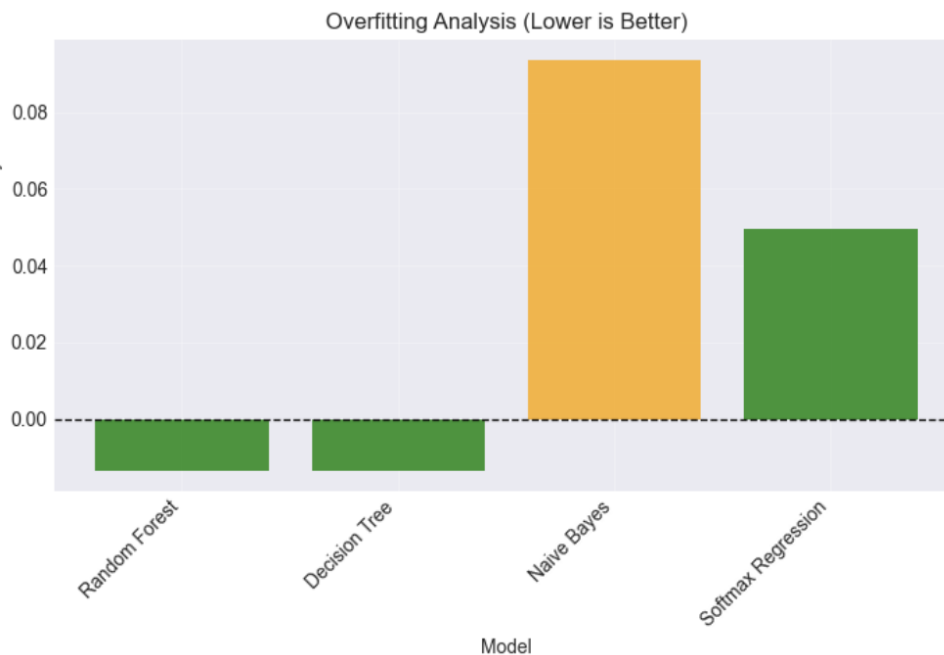


Figure 1: Overfitting Analysis (CV Score - Test Accuracy)

4.3 Metrics and Visualizations

4.3.1 Confusion Matrices

The analysis of the Confusion Matrices on the test set is fundamental to understanding the **specific error patterns** of each algorithm, especially in a context of strong class imbalance.

The cross-observation of the matrices reveals:

- **Naive Bayes (Figure 2):** The matrix has no dominant diagonal: most samples of Class 0 (*Nothing in hand*) and Class 1 are assigned to the rarer classes (4-9). This is the combined effect of the independence assumption and the balanced class weights, and it explains the 1.3% accuracy.
- **Softmax Regression (Figure 3):** Although a linear model, it shows a slight improvement in detecting some rare classes compared to Naive Bayes (thanks to the use of balanced class weight), but most errors are still concentrated in the Class 0 region.
- **Decision Tree and Random Forest (Figures 4 and 5):** These non-linear models show a significantly more structured confusion matrix. Correct classification clearly extends beyond Class 0 alone. Their errors are often confined to hierarchically close classes (e.g., misclassifying a Straight as a Pair or Two Pair), which is expected in a domain with intrinsically nuanced decision boundaries. The Random Forest presents the lowest off-diagonal counts, confirming its superiority in multi-class discrimination.

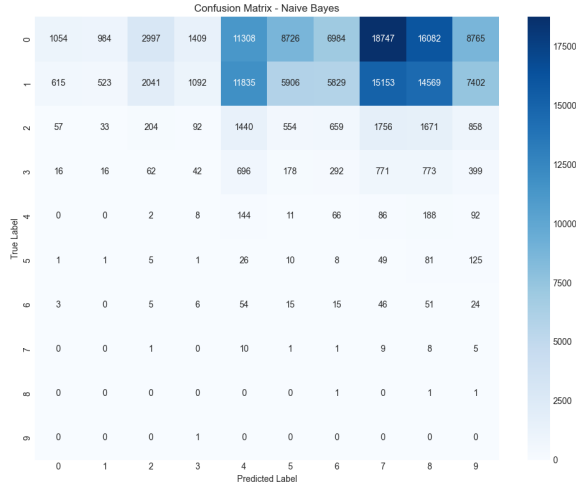


Figure 2: Confusion Matrix (Naive Bayes)

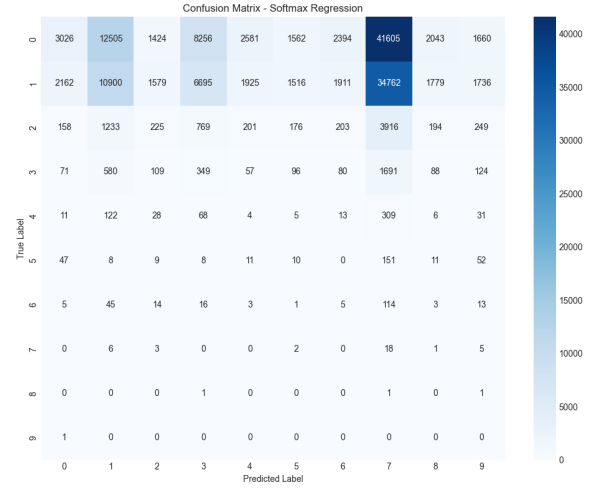


Figure 3: Confusion Matrix (Softmax Regression)

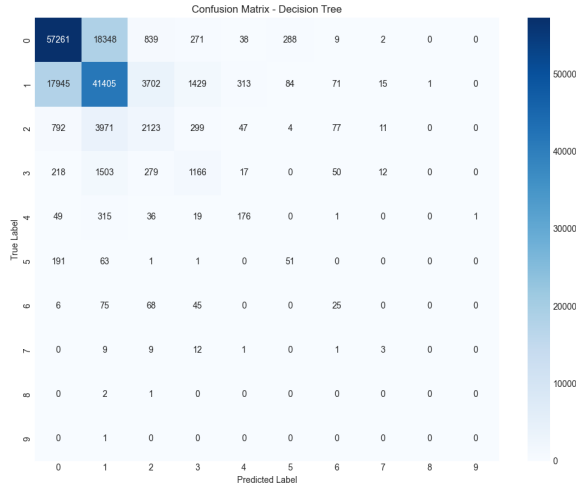


Figure 4: Confusion Matrix (Decision Tree)

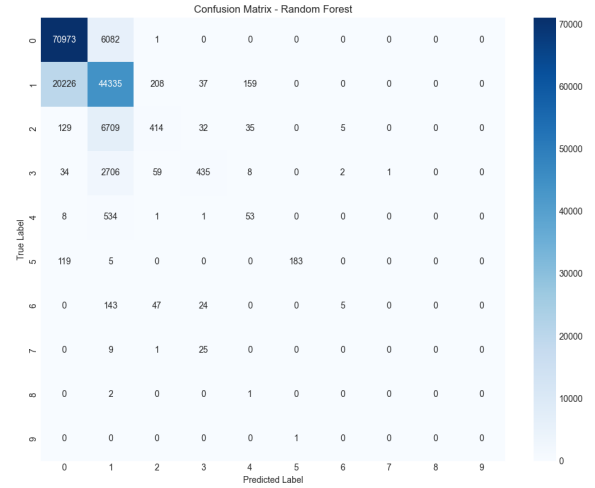


Figure 5: Confusion Matrix (Random Forest)

4.3.2 ROC and AUC

The ROC (Receiver Operating Characteristic) curves and the AUC (Area Under the Curve) metric measure the models' discrimination capacity between classes, evaluating the trade-off between True Positive Rate (Recall) and False Positive Rate. In our imbalanced multi-class context, the micro-average AUC is the most informative metric.

As observed in Figure 6, the Random Forest curve wraps around the top-left corner of the graph, reflecting the best balance between sensitivity and specificity. The micro-average AUC values for the models are:

- **Random Forest:** $AUC_{\text{micro}} = 0.9770$
- **Decision Tree:** $AUC_{\text{micro}} = 0.8138$
- **Softmax Regression:** $AUC_{\text{micro}} = 0.3704$
- **Naive Bayes:** $AUC_{\text{micro}} = 0.5876$ (Approximately the value of a random classifier, given the inability to distinguish rare classes).

These results reinforce the performance hierarchy between tree-based models and others, but present an important anomaly: the **Softmax Regression** records the lowest AUC, even lower than Naive Bayes ($AUC = 0.5876$) and the baseline value of 0.5 for random classification. This anomalous performance ($AUC = 0.3704$) suggests that the Softmax Regression model not only failed to linearly separate the

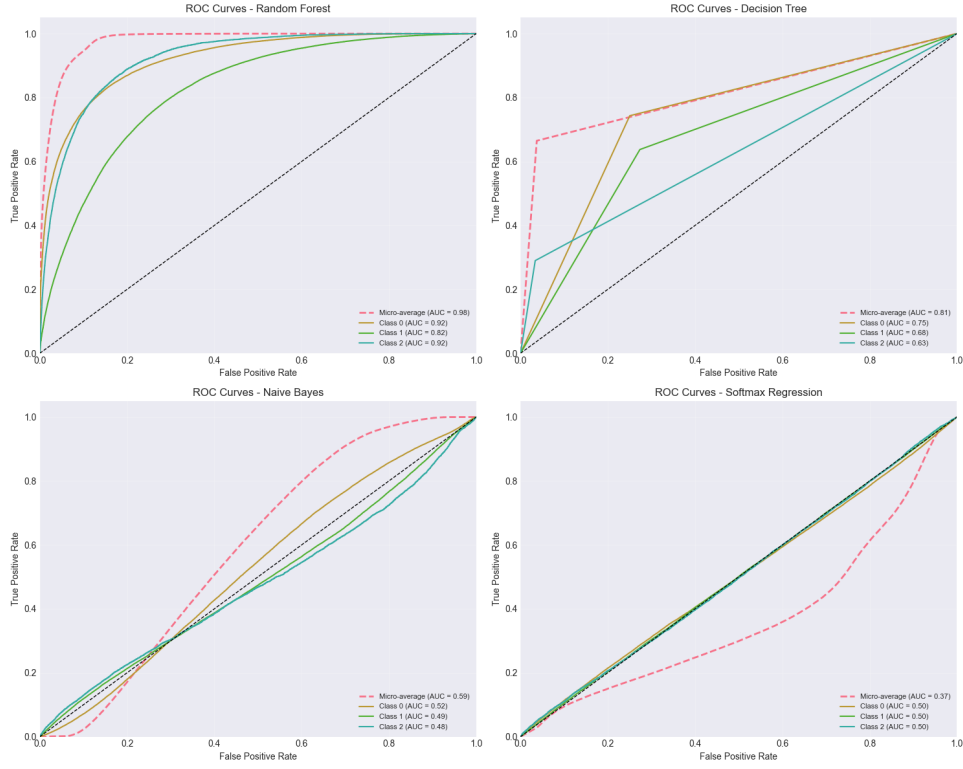


Figure 6: ROC Curves (Micro-Average) for all Compared Models.

classes but systematically assigned a higher probability to the wrong classes compared to the correct ones, despite tuning with the ‘class_weight=‘balanced’‘ option. This may stem from the extreme non-linearity of the problem, which prevented the optimization process (gradient descent) from converging to an effective discriminatory solution, getting stuck in a sub-optimal local minimum.

4.3.3 Feature Importance (Random Forest)

For tree-based models, feature importance can be extracted, which quantifies the contribution of each attribute to the reduction of impurity (or information gain) in the tree construction.

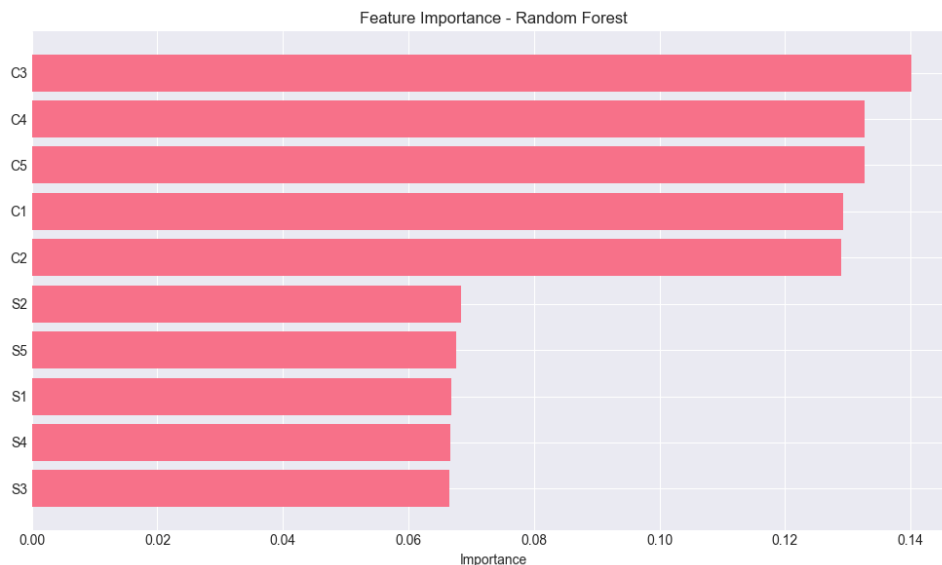


Figure 7: Feature Importance (Random Forest).

The analysis of Feature Importance (Figure 7) of the Random Forest model highlights that the **variables related to the card rank (C)** tend to have a **greater weight** than those related to the suit (S), aligning with the fact that poker hierarchies rely predominantly on rank (e.g., pairs, three of a kind) and only secondarily on suit (e.g., flush, royal flush). The values and suits of odd cards (C1, C3, S3, C5) show a marginally higher importance compared to even ones, although all 10 features contribute significantly to the decision-making process.

4.3.4 Learning Curves

Learning curves (Figure 8) analyze the trend of **bias** (error on the training set) and **variance** (gap between training and cross-validation score) as a function of the training set size.

- **Naive Bayes and Softmax Regression:** Both models show a pattern of **High Bias** (low final scores) and **Low Variance** (the two curves are very close and converge rapidly). Naive Bayes stabilizes around 0.505 accuracy, while Softmax Regression is even weaker, stabilizing below 0.15. This indicates that the models are intrinsically too simple for the dataset and that adding more data would not improve performance, confirming the inadequacy of their linearity/independence assumptions.
- **Decision Tree:** The training curve consistently maintains perfect accuracy (~ 1.0), while the validation curve stabilizes at a much lower value (~ 0.65). This wide gap is the signature of **High Variance (overfitting)**: the tree has memorized the training samples but failed to extract generalizable rules, although it improves gradually with more data.
- **Random Forest:** Similar to the Decision Tree, the Random Forest has **Low Bias** (Training Score ~ 1.0), but thanks to its ensemble nature, the Cross-validation curve is much closer and rises progressively, stabilizing at the highest value (~ 0.75). This demonstrates an excellent ability to **control variance**, overcoming the limitation of the single tree and achieving the best generalization.

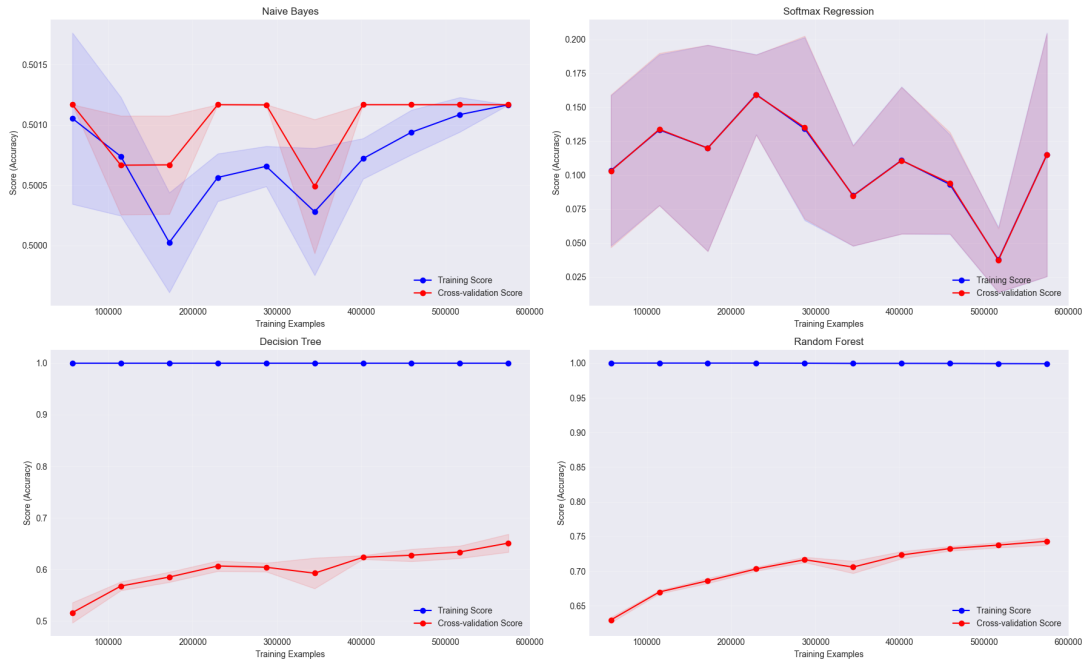


Figure 8: Learning Curves for the four models: analysis of Bias and Variance.

5 Conclusions

The project successfully achieved the goal of implementing, analyzing, and comparing four fundamental classification algorithms on the complex *Poker Hand Classification* dataset.

5.1 Summary of Results

The comparative analysis established a clear performance hierarchy among the models:

1. **Random Forest:** Best model (Accuracy 75.71%, F1 Score 0.7324, AUC_{micro} 0.9770).
2. **Decision Tree:** Good but inferior performance (Accuracy 66.48%).
3. **Naive Bayes:** Weak performance (Accuracy 1.30%), but AUC_{micro} superior to Softmax (0.5876).
4. **Softmax Regression:** Worst performance (Accuracy 9.45%, AUC_{micro} 0.3704).

5.2 Verification of Assumptions and Trade-off

The results empirically demonstrated the critical impact of model assumptions on performance:

- The **feature independence assumption** in **Naive Bayes** was drastically violated, leading to a model unusable for the discrimination of rare classes (high bias).
- The **linearity** of **Softmax Regression** proved insufficient to solve the highly non-linear decision boundaries of the poker problem, with an anomalous result suggesting the difficulty of the optimization process on this specific data representation.
- The **ensemble approach** of **Random Forest** successfully managed the **bias-variance trade-off**. While a single Decision Tree tended toward potential overfitting (high variance), the aggregation of multiple trees stabilized the model, ensuring the best generalization capacity on unseen data.

In conclusion, the **Random Forest** is the most robust model for this non-linear and imbalanced task. The project was an opportunity to learn how the choice of an algorithm must be guided not only by final accuracy but also by an understanding of how its assumptions align with the intrinsic structure of the dataset.